

FMUSER FUQ-4455A 128-IP 2-ASI Input 48-DVB-C 32-DVB-T 28-ISDB-T Non-Adjacent Carriers RF Output Mux-Modulator with 3GE-SFP for DTV Broadcasting Headend



Product Overview

The FUQ-4455A is a high-density mux-modulating device that generates up to 48 DVB-C, 32 DVB-T, and 28 ISDB-T non-adjacent carriers from 128 IP inputs via 3 GE/SFP ports and 2 GE ports, plus 32 IP inputs via 10 FE ports, with 2 ASI inputs and 1 USB Player for TS file playback. Incoming TS over UDP/RTP streams are multiplexed with PID remapping, accurate PCR adjusting, and automatic PSI/SI table generation, then modulated and output as RF across 50MHz–960MHz alongside mirrored IP (MPTS) output. The 1RU chassis delivers MER ≥ 40 dB across all modulation standards, with carrier count upgradeable by license and full control via Web-based network management. This platform is engineered for next-generation DTV broadcasting headend systems, cable operators, and terrestrial network deployments requiring multi-standard support in a compact footprint.

Key Features

Supports **128 IP inputs** via **3 GE/SFP ports** (Data1–3) and **2 GE ports** (Data4 input, Data5 bi-directional) at max **840Mbps** per channel

Accepts **32 IP inputs** via **10 FE ports** plus **2 ASI inputs** (BNC) and **1 USB Player** for TS file re-mux

Delivers **48 DVB-C**, **32 DVB-T**, or **28 ISDB-T** non-adjacent carriers with **ISDB-T/DVB-T/DVB-C** modulation dynamic switching in one device

Compliant with **ARIB STD-B31**, **ETSI EN300 744**, **DVB-C (EN 300 429)**, and **ITU-T J.83 A/B/C**; **QPSK/16QAM/64QAM/128QAM/256QAM** constellation support

Provides mirrored **IP (MPTS) output** over **UDP/RTP** unicast/multicast with **IGMP V2/V3** on Data5 bi-directional port

RF output range **50MHz–960MHz** with **1kHz stepping**, output level **-20dBm to +1dBm**, **MER ≥ 40 dB**

Per-channel **PID remapping** (auto/manual), **accurate PCR adjusting**, and automatic **PSI/SI table** generation with up to **256 PIDs** per channel

License-upgradable carrier numbers configurable before order; **Web-based network management** for system control

Inner Principle Chart

Carrier Setting Illustration:

Specifications

Input	Input	28 ISDB-T & 32 DVB-T & 48 DVB-C carriers: Max 128 IP input through Date1-Date5, 100/1000M Ethernet Port (Date1-Date3 are SFP interface optional). Each Data port can input max 128 IP. Max 32 IP input through Date1-Date10 (FE port, Rear panel), 100M Ethernet Port, Each Data port can input max 32 IP.	28 ISDB-T & 32 DVB-T & 48 DVB-C carriers: Max 128 IP input through Date1-Date5, 100/1000M Ethernet Port (Date1-Date3 are SFP interface optional). Each Data port can input max 128 IP. Max 32 IP input through Date1-Date10 (FE port, Rear panel), 100M Ethernet Port, Each Data port can input max 32 IP.
Input	Input	2ASI input, BNC interface	2ASI input, BNC interface
Input		1 USB Player input for re-mux ("xxx.ts" video)	1 USB Player input for re-mux ("xxx.ts" video)
Input	Transport Protocol	TS over UDP/RTP, unicast and multicast, IGMP V2/V3	TS over UDP/RTP, unicast and multicast, IGMP V2/V3
Input	Transmission Rate	Max 840Mbps for each input channel	Max 840Mbps for each input channel
Mux	Input Channel	128(or 32)	128(or 32)
Mux	Output Channel	28(ISDB-T) or 32 (DVB-T) or 48(DVB-C)	28(ISDB-T) or 32 (DVB-T) or 48(DVB-C)
Mux	Max PIDs	256 per channel	256 per channel
Mux	Functions	PID remapping(auto/manually optional)	PID remapping(auto/manually optional)
Mux	Functions	PCR accurate adjusting	PCR accurate adjusting
Mux	Functions	PSI/SI table automatically generating	PSI/SI table automatically generating

Modulation Parameters (ISDB-T)	Channel	16/20/24/28(License gradable)	16/20/24/28(License gradable)
Modulation Parameters (ISDB-T)	Modulation Standard	ARIB STD-B31	ARIB STD-B31
Modulation Parameters (ISDB-T)	Constellation	QPSK, 16QAM, 64QAM	QPSK, 16QAM, 64QAM
Modulation Parameters (ISDB-T)	Bandwidth	6MHz	6MHz
Modulation Parameters (ISDB-T)	Transmission Mode	2K/4K/8K	2K/4K/8K
Modulation Parameters (ISDB-T)	Code rate	1/2, 2/3, 3/4, 5/6, 7/8	1/2, 2/3, 3/4, 5/6, 7/8
RF Output (ISDB-T)	Interface	F typed output port for 28 non-adjacent carriers	F typed output port for 28 non-adjacent carriers
RF Output (ISDB-T)	RF Range	50~960MHz, 1kHz stepping	50~960MHz, 1kHz stepping
RF Output (ISDB-T)	Output Level	-20dBm~1dBm(87~108dBμV), 0.1dB stepping	-20dBm~1dBm(87~108dBμV), 0.1dB stepping
RF Output (ISDB-T)	MER	≥ 40dB	≥ 40dB
RF Output (ISDB-T)	ACL	-50 dBc	-50 dBc
Modulation Parameters (DVB-T)	Channel	16/20/24/28/32(License gradable)	16/20/24/28/32(License gradable)
Modulation Parameters (DVB-T)	Modulation Standard	ETSI EN300 744	ETSI EN300 744
Modulation Parameters (DVB-T)	Constellation	QPSK/16QAM/64QAM	QPSK/16QAM/64QAM
Modulation Parameters (DVB-T)	Bandwidth	6/7/8 MHz	6/7/8 MHz
Modulation Parameters (DVB-T)	Transmission Mode	2K/4K/8K	2K/4K/8K
Modulation Parameters (DVB-T)	Code rate	1/2,2/3,3/4,5/6,7/8	1/2,2/3,3/4,5/6,7/8
RF Output (DVB-T)	Interface	F typed output port for 32 non-adjacent carriers	F typed output port for 32 non-adjacent carriers

RF Output (DVB-T)	RF Range	50~960MHz, 1kHz stepping	50~960MHz, 1kHz stepping
RF Output (DVB-T)	Output Level	-20dBm~0dBm(87~107dbμV), 0.1dB stepping	-20dBm~0dBm(87~107dbμV), 0.1dB stepping
RF Output (DVB-T)	MER	≥ 40dB	≥ 40dB
RF Output (DVB-T)	ACL	-55 dBc	-55 dBc
Modulation Parameters (DVB-C)	Constellation	J.83A	Constellation :16/32/64/128/256QAM
Modulation Parameters (DVB-C)	Constellation	J.83A	Bandwidth :8M
Modulation Parameters (DVB-C)	Constellation	J.83B/C	Constellation :64/256QAM
Modulation Parameters (DVB-C)	Constellation	J.83B/C	Bandwidth :6M
Modulation Parameters (DVB-C)	QAM Channel	32/36/40/44/48(License gradable)	32/36/40/44/48(License gradable)
Modulation Parameters (DVB-C)	Symbol Rate	5.0~7.0Msps, 1ksps stepping. 5057Ksps (J.83B, 64QAM) ; 5361Ksps (J.83B, 256QAM)	5.0~7.0Msps, 1ksps stepping. 5057Ksps (J.83B, 64QAM) ; 5361Ksps (J.83B, 256QAM)
Modulation Parameters (DVB-C)	Constellation	16, 32, 64, 128, 256QAM	16, 32, 64, 128, 256QAM
RF Output (DVB-C)	Interface	F typed output port for 48 non-adjacent carriers	F typed output port for 48 non-adjacent carriers
RF Output (DVB-C)	RF Range	50~960MHz, 1kHz stepping	50~960MHz, 1kHz stepping
RF Output (DVB-C)	Output Level	-20dBm~-3dBm(87~104dbμV), 0.1dB stepping	-20dBm~-3dBm(87~104dbμV), 0.1dB stepping
RF Output (DVB-C)	MER	≥ 40dB	≥ 40dB
RF Output (DVB-C)	ACL	50~960MHz, 1kHz stepping	50~960MHz, 1kHz stepping

TS output	28&32&48 IP output over UDP/RTP, unicast/multicast, Data1 100/1000M Ethernet Ports	28&32&48 IP output over UDP/RTP, unicast/multicast, Data1 100/1000M Ethernet Ports	28&32&48 IP output over UDP/RTP, unicast/multicast, Data1 100/1000M Ethernet Ports
System	Web-based Network management	Web-based Network management	Web-based Network management
General	Demission	420mm×440mm×44.5 mm (W×L×H)	420mm×440mm×44.5 mm (W×L×H)
General	Temperature	0~45■(operation), -20~80■(storage)	0~45■(operation), -20~80■(storage)
General	Power Supply	AC100V±10%, 50/60Hz or AC 220V±10%,50/60Hz	AC100V±10%, 50/60Hz or AC 220V±10%,50/60Hz

Order Guide

Modulation combination	DVB-C/DVB-T	DVB-C/ISDB-T	DVB-T/ISDB-T
Carrier support (Increase every four carriers)	32~48/16~32	32~48/16~28	16~32/16~28

Modulation	Carrier number(Increase every four carriers by license)	Carrier number(Increase every four carriers by license)	Carrier number(Increase every four carriers by license)	Carrier number(Increase every four carriers by license)	Carrier number(Increase every four carriers by license)
DVB-C	32	36	40	44	48
DVB-T	16	20	24	28	32
ISDB-T	16	20	24	28	32
Note:	Different combination of carrier number can be selected, Need to be confirmed before an order	Different combination of carrier number can be selected, Need to be confirmed before an order	Different combination of carrier number can be selected, Need to be confirmed before an order	Different combination of carrier number can be selected, Need to be confirmed before an order	Different combination of carrier number can be selected, Need to be confirmed before an order

About Us

FMUSER is a company specializing in broadcasting and television transmission systems and integrating hotel IPTV systems. The company was established in 2009 and has served over 180 countries worldwide. More than 20,000 television stations, radio stations, and hotel clients from different.

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